Lecture 11: Gas & Ice Giants
Jupiter, Saturn, Uranus, Neptune

Planetary Systems

Tim Lichtenberg

Kapteyn Astronomical Institute
University of Groningen

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Jupiter’s south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS

Learning Objectives

By the end of this lecture, you will be able to:

- Compare the interiors of the four giant planets
- Describe the atmospheric dynamics of Jupiter, Saturn, Uranus and Neptune
- Outline the diversity of giant-planet satellites and rings
- Derive the Roche limit and apply it to Saturn's rings
- Use the ice giants as the closest solar-system analogues of the commonest exoplanets

Recap: Where We Are in the Course

- **L9 (Earth & Venus):** rocky-planet near-twins with divergent surface evolutions
- **L10 (Mercury & Mars):** rocky-planet limiting cases (dense core vs. volatile-rich)
- **Today:** the four giant planets, >99.5% of all planetary mass beyond the Sun

Today: How do gas and ice giants work, and why do their moons host oceans?

1. Gas Giants and Ice Giants: An Overview

Solar system family portrait, Voyager 1. Credit: NASA/JPL-Caltech

Two Classes, Four Planets

- **Gas giants** (Jupiter, Saturn):
  - H/He envelopes dominate mass (318 and 95 $M_\oplus$)
  - Together: ~ 92% of all planetary mass
- **Ice giants** (Uranus, Neptune):
  - H/He only ~ 10–20% of mass (14.5 and 17.1 $M_\oplus$)
  - Bulk = water, ammonia, methane fluids

- No solid surface; “1 bar level” is standard reference
- Smooth gas → supercritical fluid → metallic plasma
- Rapid rotation (~ 10 h), oblate by 6–10%

Energy Budgets: Internal Heat Excess

Three of four giant planets radiate far more energy than they receive from the Sun; Uranus is the anomalously cold exception.

- **Jupiter (**1.7×**):** slow Kelvin-Helmholtz gravitational contraction
- **Saturn (**1.8×**):** gravitational energy release from helium rain
- **Neptune (**2.6×**) vs. Uranus (**1.06×**):** Neptune has strong internal flux; Uranus is near solar equilibrium

Why Rapid Rotation Matters

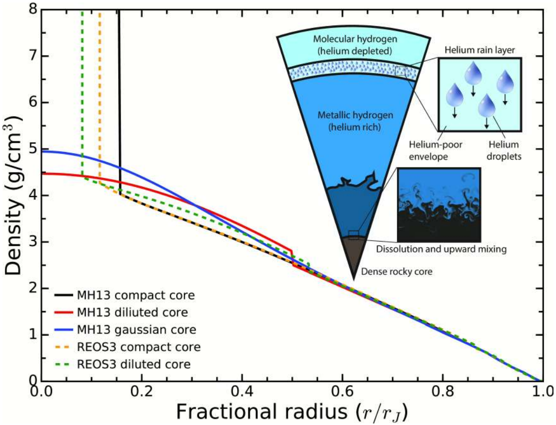
Rapid rotation drives planetary oblateness, latitude-aligned zonal winds, and gravitational harmonics that reveal the deep density profile.

- **Rapid spin:** Jupiter (9 h 56 min) and Saturn (10 h 33 min) rotate in ~ 10 h
- **Zonal banding:** powerful Coriolis forces organise atmospheric circulation into alternating jets
- **Gravity harmonics J_n:** oblateness and deep flows encode mass distribution, measured by Juno and Cassini

2. Jupiter: Interior and Atmosphere

Jupiter’s south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS

Jupiter Interior: Layered Fluid



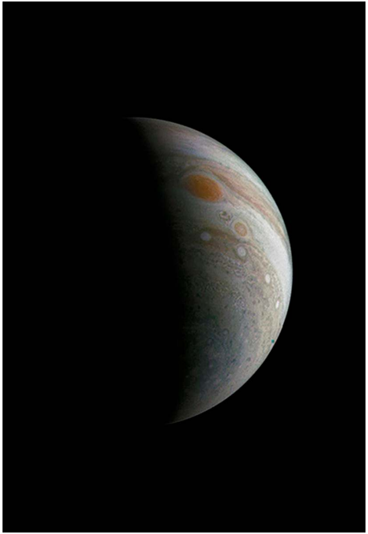
Source: Wahl et al. (2017)

- Outer molecular H_2 + He envelope
- Transition to metallic hydrogen at ~ 100 GPa, ~ 5000 K
- $r/R_J \approx 0.85$ for the conducting transition
- Dynamo source (field-generating fluid motion): metallic hydrogen
- Centre: ~ 4000 GPa, ~ 20,000 K

The Dilute Core: A Decade-Old Revolution

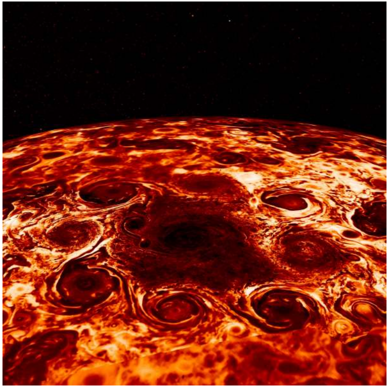
Juno gravity measurements revealed that Jupiter does not possess a compact core, but rather an extended, dilute distribution of heavy elements.

- **Gravity harmonics:** J_4–J_{10} cannot match compact ~ 10 $M_\oplus$ core models
- **Dilute core:** 20–40 $M_\oplus$ of heavy elements distributed across inner 30–50% of radius
- **Formation rewrite:** the dilute core (Wahl et al. 2017; Militzer et al. 2022) implies efficient mixing during accretion or a giant impact (Liu et al. 2019)

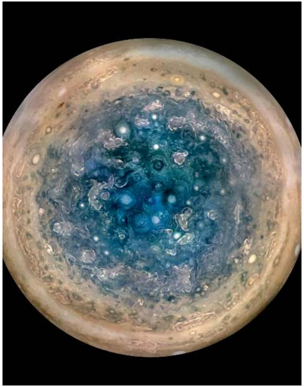


Crescent Jupiter and Great Red Spot, JunoCam (perijove 3, Dec 2016). Alternating zonal cloud belts bracket the anticyclonic Great Red Spot, which has shrunk from ~ 40,000 km in 1900 to ~ 14,000 km today. Credit: NASA/JPL-Caltech/SwRI/MSSS / R. Tkachenko.

Polar Cyclones



North pole: 1 + 8 cyclones, Adriani et al. (2018)



South pole: 1 + 5 cyclones

Stable polygonal arrangements persist across many Juno orbits. Polar weather not seen at high resolution before 2016.

Jets, Aurorae, and the Great Blue Spot

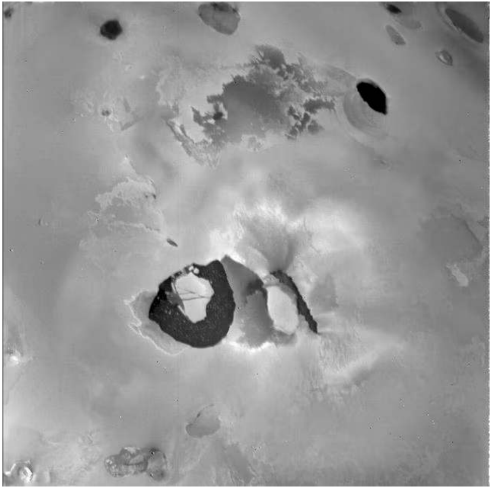
Jets reach ~ 3000 km deep; Io feeds the strongest aurorae in the solar system.

- **Jet penetration:** winds extend to ~ 3000 km before magnetic braking enforces rigid rotation (Kaspi et al. 2018)
- **Io mass loading:** ~ 1 $\mathrm{t\,s^{-1}}$ of sulfur and oxygen ions enter the magnetosphere
- **Auroral footprints:** UV spots mark the field lines to Io, Europa and Ganymede
- **Great Blue Spot:** isolated equatorial magnetic anomaly mapped by Juno (Connerney et al. 2022)

3. The Galilean Moons

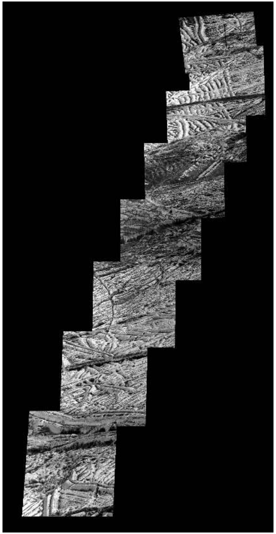
The Galilean moons at correct relative size: Io, Europa, Ganymede, Callisto. Credit: NASA/JPL/DLR

Io: Most Volcanic Body in the Solar System



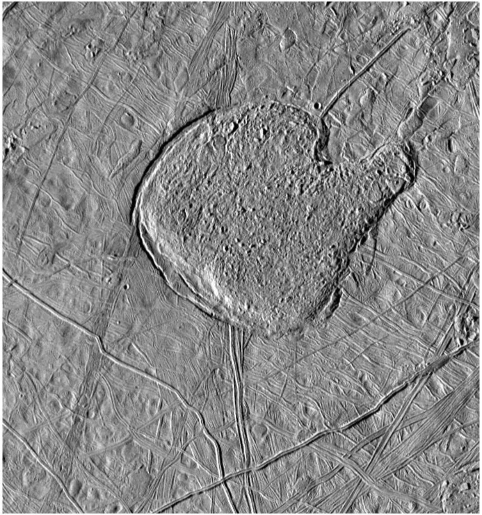
Loki Patera lava lake, Voyager 1 (1979). NASA/JPL

- ~ 400 active volcanic centres
- Heat output ~ 10^{14} W (tidal, not radiogenic)
- Driven by 1:2:4 Laplace resonance with Europa and Ganymede
- Predicted by Peale et al. (1979) days before Voyager 1
- Surface resurfaced on ~Myr timescales



Europa surface mosaic from the Galileo orbiter. Global lineae fractures and disrupted chaos terrain reflect ongoing tidal heating that yields a young surface aged only 40–90 Myr. Credit: NASA/JPL-Caltech/SETI Institute.

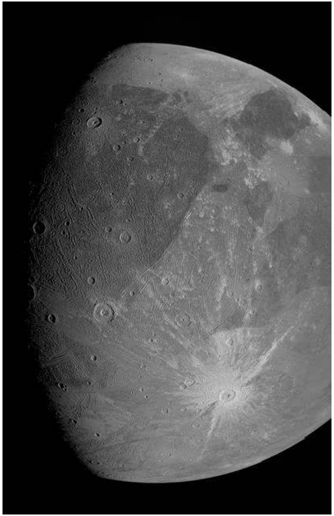
The Case for Europa's Ocean



Conamara Chaos, Galileo PIA01640

- Chaos morphology → episodic basal melting
- Galileo magnetometer: induced dipole requires global conducting ocean (Khurana et al. 1998)
- Hubble UV: candidate plumes near south pole (Roth et al. 2014)
- Modern picture: 6–25 km ice over ~ 100 km ocean (~Earth-ocean volume)

Ganymede: Largest Moon, Only Moon-Dynamo

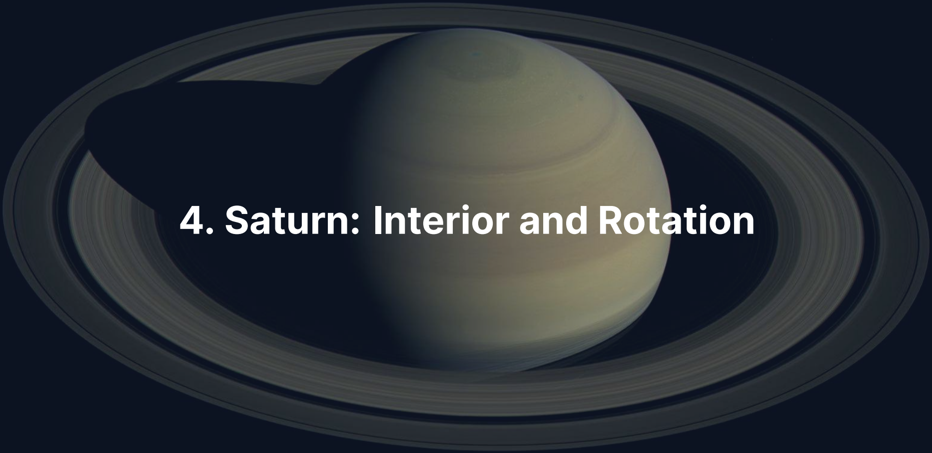


- Radius 2634 km: larger than Mercury
- Only moon known to host an intrinsic dynamo (Galileo 1996)
- Fully differentiated: iron core → silicate mantle → icy shell
- Deep ocean: ~ 100 km thick below ~ 150 km ice shell (Saur et al. 2015)

Callisto: The Quiet Sibling

- R = 2410 km, density 1834 $\mathrm{kg\,m^{-3}}$
- $C/MR^2 \approx 0.355$: only partially differentiated (Anderson et al. 2001)
- No tectonic resurfacing; sits outside 1:2:4 Laplace resonance
- Induced magnetic field signal indicates probable subsurface ocean
- Lower radiation environment than inner Galilean moons

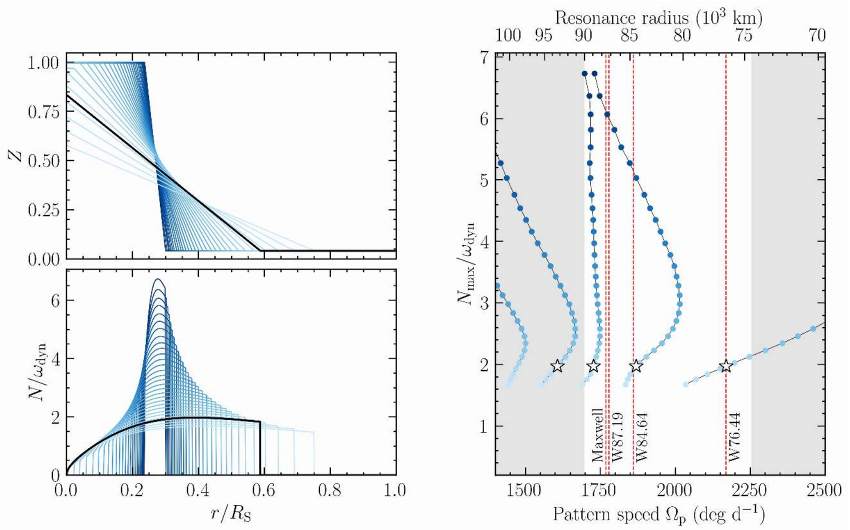
4. Saturn: Interior and Rotation



Saturn and its rings, Cassini (PIA21046). Credit: NASA/JPL-Caltech/SSI

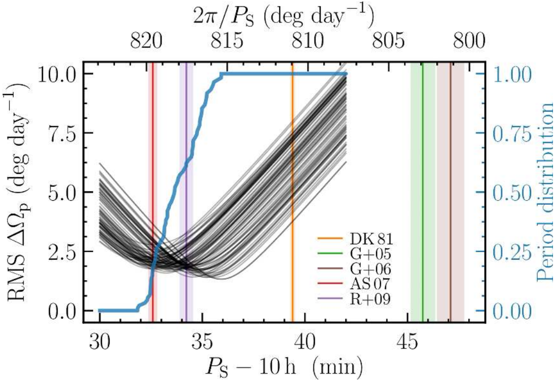
Saturn Interior: Lower Pressures, Helium Rain

- Mass 95 $M_\oplus$; central pressure ~ 1/3 of Jupiter
- Molecular-to-metallic hydrogen transition occurs deeper and at milder conditions
- Cassini Grand Finale (2017) mapped high-precision gravity field
- Helium becomes immiscible in metallic H at 1–3 Mbar, 5–10 kK
- Falling droplets release gravitational energy, closing Saturn's heat budget
- Depletes upper-envelope helium, consistent with Voyager atmospheric observations



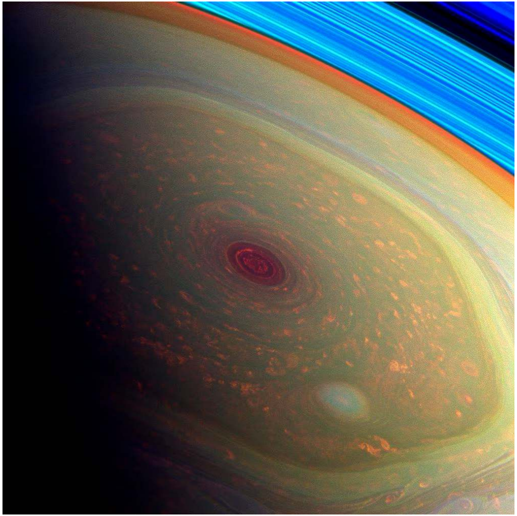
Kronoseismology constraints on Saturn's interior: f-mode resonances with C-ring waves require a stably stratified dilute core out to ~ 60% of R_S with ~ 17 $M_\oplus$ of heavy elements. Mankovich & Fuller (2021).

Saturn's Rotation Period from Ring Seismology



- Saturn's magnetic dipole is nearly axisymmetric, precluding a radio period
- C-ring f-mode wave frequencies provide seismological clock
- Best-fit rotation period: P_S = 10 h 33 min 38 s
- Resolves decades-long Voyager-Cassini rotation ambiguity (Mankovich et al. 2019)

Saturn's Atmosphere and the Hexagon



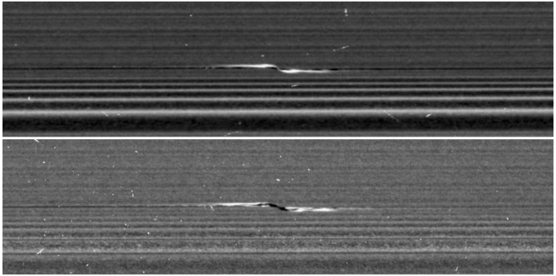
- Same $\mathrm{NH_3}$/$\mathrm{NH_4SH}$/$\mathrm{H_2O}$ cloud decks as Jupiter, deeper and muted
- Hexagonal jet at ~ 78° N: persistent six-sided standing Rossby wave
- Observed since the Voyager flybys (1980 to 1981)
- Equatorial jet reaches ~ 400 $\mathrm{m\,s^{-1}}$ (Cassini)

5. Saturn’s Rings

Saturn’s main ring system, Cassini (PIA21046). Credit: NASA/JPL-Caltech/SSI

Natural-colour radial scan across the C ring, B ring, Cassini Division, and A ring; the rings span 67,000–140,000 km, consist of > 95% pure water ice, and are only ~ 10 m thick. Cassini (PIA08389). Credit: NASA/JPL-Caltech/SSI.

Embedded Moons and Propellers



- F ring shepherded gravitationally by Prometheus and Pandora
- Encke Gap cleared by embedded moon Pan
- Propeller features: ~ 1 km moonlets embedded in the A ring
- Direct detection of orbital migration within a disk (Tiscareno et al. 2013)

Ring Youth and Origin

Saturn's rings are a transient feature with an estimated lifetime of ~ 10^8 yr.

- **Low mass:** ~ 1.5×10^{19} kg, ~ 40% of Mimas (Iess et al. 2019)
- **Not primordial:** such light rings would have darkened by micrometeoroid infall over 4.5 Gyr; their brightness gives ~ 100 Myr (Crida et al. 2019)
- **Ring rain:** water drains into Saturn's atmosphere, depleting rings on 10^8 yr scales
- **Disruption origin:** formed by tidal disruption of an icy moon (Wisdom et al. 2022)

6. Blackboard Derivation: The Roche Limit

Solar system family portrait. Credit: NASA/JPL-Caltech

Setup: Why Rings Exist

- **Roche limit:** distance inside which tidal stretch overcomes self-gravity (Roche 1849)
- Sets where rings can persist and where moons cannot accrete

Configuration:

- Planet: mass M_p, radius R_p, mean density ρ_p
- Satellite: mass M_s, radius R_s, mean density ρ_s, at orbital distance d
- Assume $d \gg R_s$ for differential tidal expansion across the satellite

Balance the tidal stretch $\Delta a_{\mathrm{tidal}} \approx 2GM_pR_s/d^3$ against the surface self-gravity GM_s/R_s^2. *To the board.*

Fluid vs. Rigid Roche Limit

- Rigid case: the body holds its spherical shape until disruption: $d_R^{\text{rigid}} = 2^{1/3} R_p (\rho_p / \rho_s)^{1/3} \approx 1.26\, R_p (\rho_p / \rho_s)^{1/3}$
- Fluid case: the body deforms into an ellipsoid first and disrupts further out; Roche's calculation gives the prefactor ≈ 2.46:

$$\boxed{d_R \approx 2.46\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}}$$

Real bodies with material strength sit between the rigid and fluid limits.

Apply to Saturn's Rings

$$R_p = 58{,}232\ \mathrm{km}, \quad \rho_p = 687\ \mathrm{kg\,m^{-3}}, \quad \rho_s \approx 1000\ \mathrm{kg\,m^{-3}}$$

$$d_R \approx 2.46 \times 58{,}232 \times (0.687)^{1/3}\ \mathrm{km} \approx 2.46 \times 58{,}232 \times 0.883$$

$$\boxed{d_R \approx 126{,}000\ \mathrm{km}}$$

Within ~ 10% of A-ring outer edge (~ 137,000 km). Discrepancy reflects particle porosity, ice rigidity, and resonant structures.

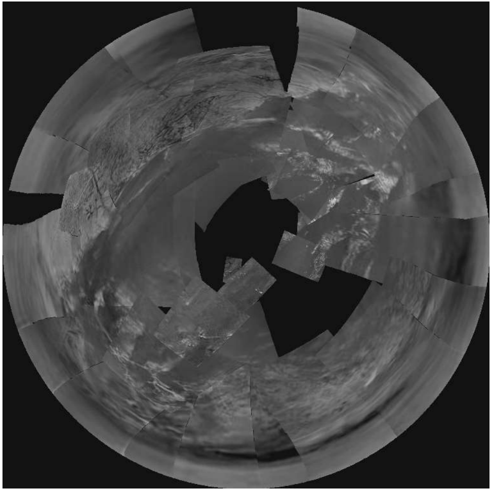
Break

Earthrise, Apollo 8 (1968). Credit: NASA

7. Saturn’s Moons: Titan, Enceladus, and More

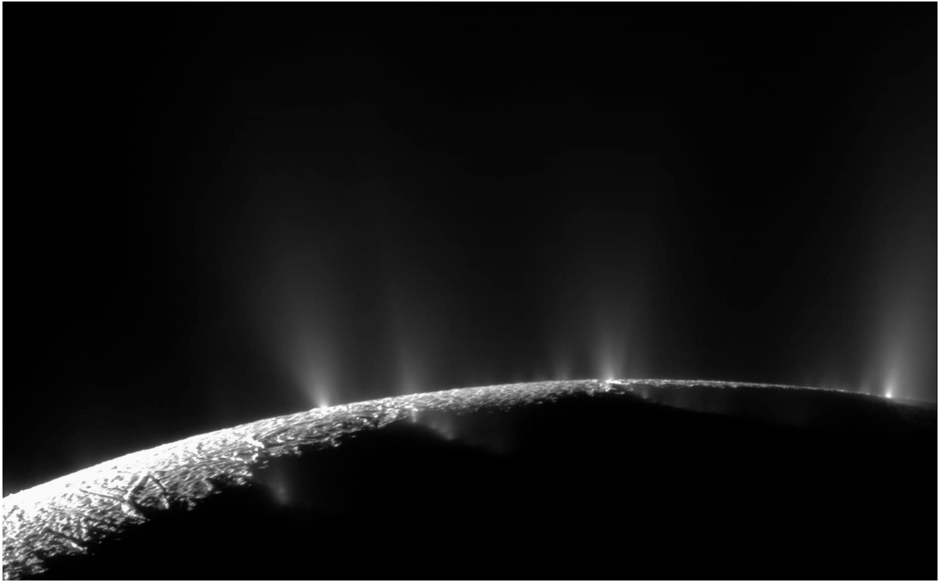
Titan’s polar lakes, Cassini RADAR. NASA/JPL-Caltech/ASI/Cornell

Titan: Atmosphere and Methane Cycle



Huygens descent imagery, 14 Jan 2005 (PIA07870)

- R = 2575 km; second largest moon in the solar system
- N_2 atmosphere: 1.5 bar surface, 4× Earth column density
- Surface 94 K: methane sits near its triple point
- Methane hydrological cycle: clouds, rain, rivers, and polar hydrocarbon lakes



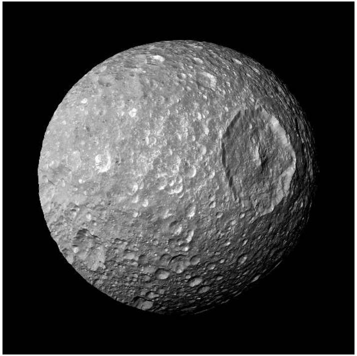
Dozens of cryovolcanic geyser jets erupt from active south-polar tiger stripe fractures powered by > 10 GW of tidal heat. Cassini, NASA/JPL-Caltech/SSI.

Enceladus: Habitability Ingredients

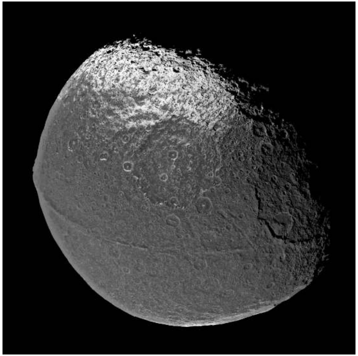
- Global subsurface ocean beneath a 20–30 km ice shell; the 2:1 resonance with Dione forces the eccentricity that powers the tidal heating
- Plume $\mathrm{H_2}$ (Waite et al. 2017): chemical disequilibrium from hydrothermal vents
- Silica nanoparticles (Hsu et al. 2015): active rock-water reactions
- Phosphates (Postberg et al. 2023): essential biochemical building block

> Most accessible ocean world: plumes spray subsurface ocean water directly into space.

Mid-Sized Moons: Mimas and Iapetus

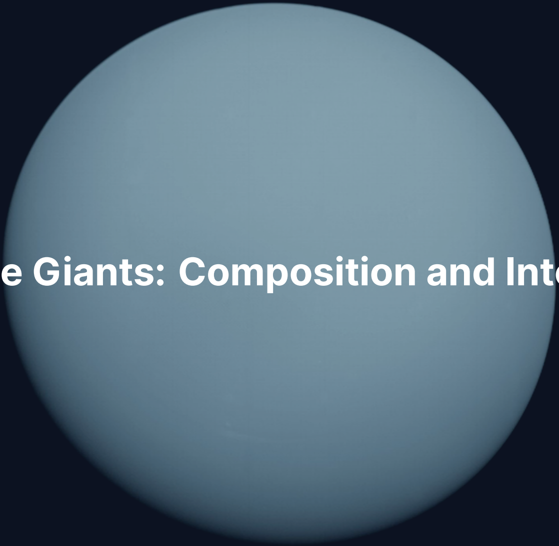


Mimas, Herschel crater (near disruption threshold)



Iapetus, two-toned hemispheres

8. Ice Giants: Composition and Interior



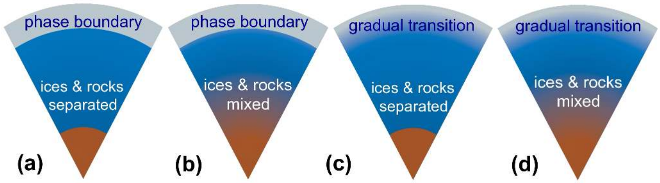
Uranus, Voyager 2 (1986). NASA/JPL-Caltech

Ice Giants: The Exotic Twins

- Uranus: 14.5 $M_\oplus$, 4.0 $R_\oplus$; Neptune: 17.1 $M_\oplus$, 3.9 $R_\oplus$
- H/He envelope only 10–20% of total mass
- Bulk composition: water, ammonia, and methane in compressed fluid state
- One spacecraft visit each: Voyager 2 in Jan 1986 and Aug 1989

Most under-explored major planets; data are 35–40 yr old.

Possible Interior Structures



- (a) Sharp boundaries
- (b) Sharp envelope, gradual core
- (c) Gradual envelope, sharp core
- (d) Fully gradual gradient
- Voyager J_2, J_4 cannot break the density profile degeneracy
- Helled et al. (2020)

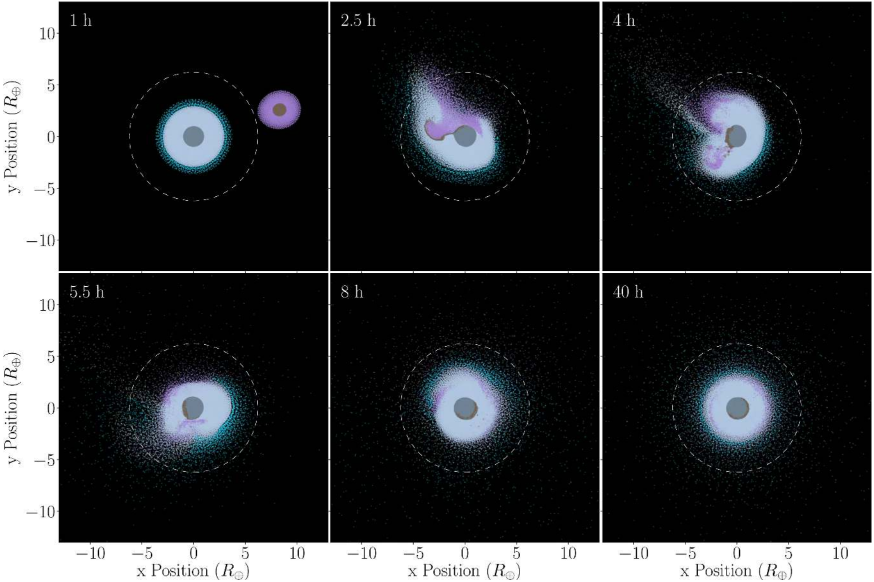
Superionic Ice and Tilted Dynamos

- **Superionic ice:** at 100–400 GPa a solid oxygen lattice with mobile protons conducts (Millot et al. 2019)
- **Tilted, offset fields:** 59° (Uranus) and 47° (Neptune), offset ~ $1/3\,R_p$ and ~ $1/2\,R_p$ from the centre
- **Thin-shell dynamos:** convection in an outer shell explains the multipolar geometry

Superionic ice powers dynamos without metallic hydrogen.

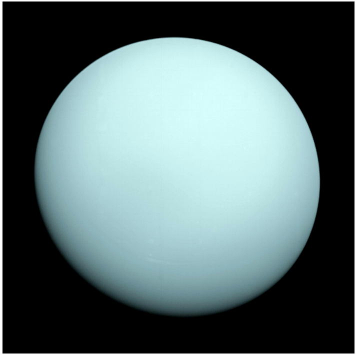
9. Uranus: The Tilted Planet

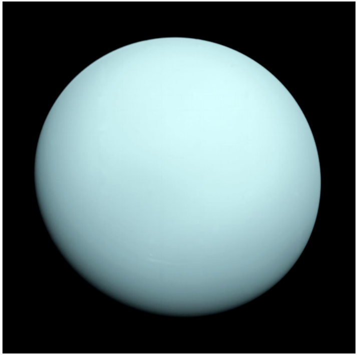
Uranus clouds, Voyager 2 (PIA18182)



SPH simulation of a giant impact on proto-Uranus: an oblique collision tilted the spin axis to 97.8° (yielding 42-yr polar seasons) and shed an equatorial debris disk. Kegerreis et al. (2018).

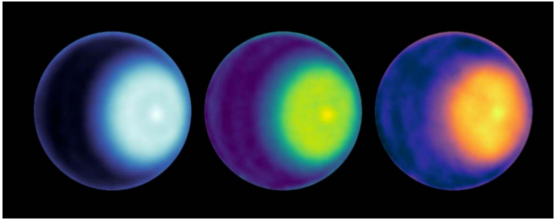
Uranus: Featureless at Voyager, Active Now





Voyager 2 (1986) observed Uranus near solstice: pole-on orientation and muted cloud bands.

Modern Uranus: Storms and Cyclones



- Since the 2007 equinox, toward northern summer (2030): cloud activity and storms increase
- 2014 storm system detected by ground-based observers
- JWST (2023): vivid rings and polar cap structure
- North-polar cyclone confirmed by VLA and JWST

Uranus: Why So Cold Inside?

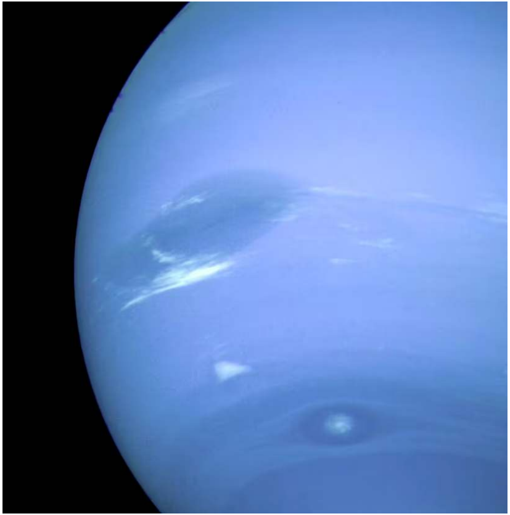
Uranus emits almost no heat beyond the sunlight it absorbs; Neptune does.

- **Heat anomaly:** ~ 1.06× the absorbed solar flux (Pearl et al. 1990); Neptune ~ 2.6×
- **Trapped heat:** a giant impact may have left a stably stratified interior that suppresses convection
- **Latent heat:** deep condensation of volatiles could delay the cooling

10. Neptune and Triton

Neptune Great Dark Spot, Voyager 2 (1989). NASA/JPL-Caltech

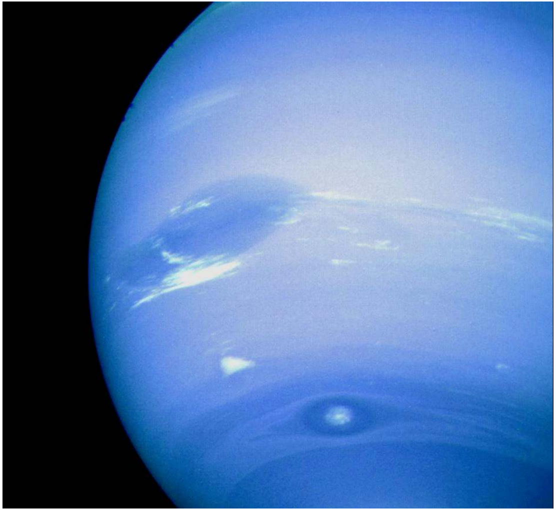
Neptune: The Active Ice Giant



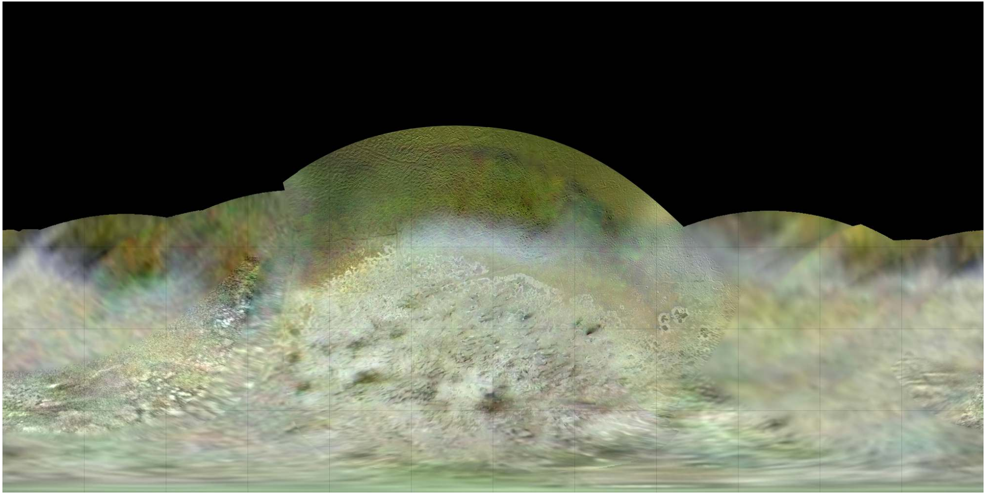
Great Dark Spot, Voyager 2 (1989)

- Active anticyclonic storm observed during the 1989 flyby
- Great Dark Spot was comparable in size to Earth
- Bordered by bright methane cirrus clouds
- Disappeared by 1994 (HST); new dark spots emerge periodically

Neptune: Fastest Winds in the Solar System



- “Scooter”: bright cloud feature near 42°S, drifting westward relative to the 16.1 h interior rotation
- Equatorial easterly jet peaks at ~ 400 $\mathrm{m\,s^{-1}}$
- Insolation is only 1/900 of Earth’s: wind energy must derive from the interior
- Internal heat excess: emits ~ 2.6× absorbed solar flux (Pearl & Conrath 1991)



Triton's southern hemisphere, Voyager 2 mosaic: a captured Kuiper Belt object: regular moons form prograde in the circumplanetary disc, so its retrograde orbit ($i \approx 157^{\circ}$) means capture; it displays active 8-km $\mathrm{N_2}$ geysers, cantaloupe terrain, and a young surface aged ≲ 100 Myr. Credit: NASA/JPL-Caltech.

11. Ice Giant Rings

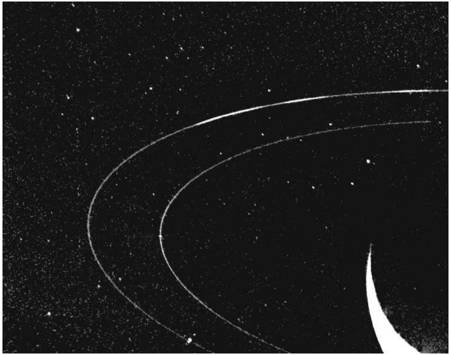
Neptune ring system, Voyager 2 (1989). NASA/JPL-Caltech

Ice Giant Rings: Narrow, Dark, and Carbonaceous

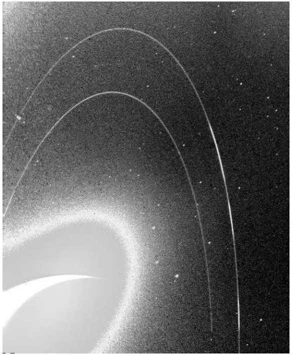
Ice-giant rings are narrow, dark, and carbon-rich, contrasting sharply with Saturn's broad water-ice sheet.

- **Occultation discovery:** Uranus's rings were discovered in 1977 via stellar occultation (13 narrow rings known today)
- **Compositional contrast:** Saturn's rings are > 95% bright water ice, whereas ice-giant rings are dark organics
- **Disruption origin:** sourced by collisional destruction of small inner moonlets rather than a single large icy parent

Neptune Rings: Arcs Held by Resonance



Adams ring + Le Verrier ring



Long-exposure forward-scattered view

Adams ring contains 5 distinct *arcs*, gravitationally trapped by mean-motion resonance with Galatea.

12. Comparative Payoff and Exploration Frontier

Europa Clipper concept. NASA/JPL-Caltech

Why Gas Giants and Ice Giants Diverged

- **Timing:** Jupiter and Saturn reached runaway gas accretion while the nebula was still massive; Uranus and Neptune grew too slowly and caught only 1–4 $M_\oplus$ of H/He
- **Result:** two H/He-dominated planets with metallic hydrogen dynamos, two ice-dominated planets with superionic-ice dynamos
- **Common threads:** rapid rotation, internal heat (except Uranus), rings and regular moon systems, one family of physics

Europa Clipper (launched 2024, ~ 50 Europa flybys) and JUICE (launched 2023, Ganymede orbit 2034) are en route to investigate the habitability of Jovian ocean worlds. Credit: NASA/JPL-Caltech.

Summary: Today's Course-Level Takeaways

1. **Two classes:** gas giants (H/He-dominated, Saturn ~ 80% H/He by mass) vs. ice giants (H/He only 10–20%)
2. **Dilute cores:** Juno and ring seismology reveal diffuse cores in Jupiter and Saturn
3. **Roche limit:** $d_R \approx 2.46\, R_p(\rho_p/\rho_s)^{1/3}$ bounds rings and moon accretion
4. **Ocean moons:** tidal heating maintains oceans within Europa and Enceladus
5. **Under-explored ice giants:** unique tilted dynamos powered by superionic ice, and the closest analogues of the sub-Neptunes of Lecture 13

Next: Lecture 12

Questions?

Next: **Lecture 12. Small Bodies: Meteorites, Asteroids, Comets**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`